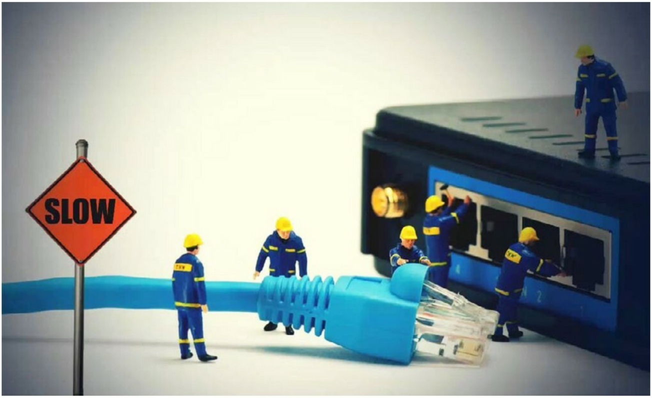
iptables: The Default Linux Firewall

This article deals with iptables, which is a built-in firewall in Linux. The authors explain the commands to configure iptables for various situations, thus making this a must-read for newbies.



The term 'firewall' generally refers to a barrier that is used to limit the spread of fire. In the computing world, it refers to a software or hardware based network security system, which can be used to control incoming and outgoing network traffic based on a set of rules.

A firewall basically establishes a barrier between the internal network (a group of systems or a single one), which is assumed to be secure and trusted, and the external network (usually the Internet), which is considered neither secure nor trusted. Various operating systems include software based firewalls to protect against the threats from the Internet. A router also consists of firewalls, and a firewall can also perform routing functions.

Figure 2 shows the generation of a firewall, while Figure 3 lists the types of firewalls.

iptables

iptables is a built-in firewall in Linux. It is a user based application for configuring the tables provided by the Linux kernel firewall. iptables is the default firewall installed with Red Hat, CentOS, Fedora Linux, etc. Different modules and programs are used for different protocols such as iptables for IPv4, ip6tables for IPv6 and so on. It uses the concept of IP addresses, protocols (tcp, udp, icmp, etc) and ports.

iptables is a command line firewall that uses the concept of chains to handle the network traffic. It places the rules into chains, i.e., INPUT, OUTPUT and FORWARD, which are checked against the network traffic. Decisions are made as to what to do with the packets based on these rules, i.e., whether the packet should be accepted or dropped. These actions are referred to as targets. DROP and ACCEPT are commonly used predefined targets used for dropping and accepting the packets, respectively.

The three predefined chains in the filter table to which rules are added for processing IP packets are:

INPUT: These are packets destined for the host computer.

OUTPUT: These are packets originating from the host computer.

FORWARD: These packets are neither destined for nor originate from the host computer, but pass through (routed by) the host computer. This chain is used if you are using your computer as a router.